INCH-POUND

NOTE The document identifier and heading have been changed on this page to reflect that this is a performance specification There are no other changes to this document The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised

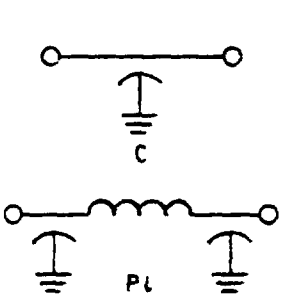
MIL-PRF-28861/10E
23 August 1993
SUPERSEDING
MIL-F-28861/10D
2 November 1988

PERFORMANCE SPECIFICATION SHEET

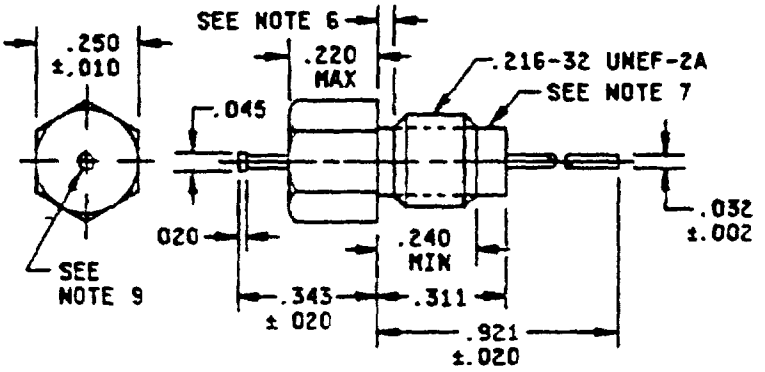
FILTERS AND CAPACITORS, RADIO FREQUENCY/ELECTROMAGNETIC INTERFERENCE SUPPRESSION, HERMETICALLY SEALED, STYLE FS64

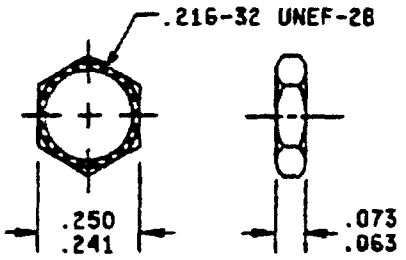
This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-F-28861

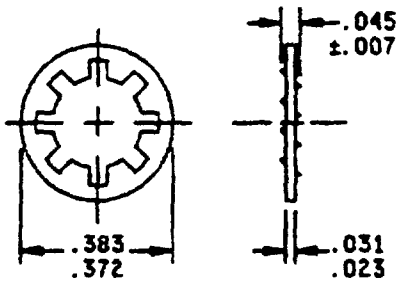


CIRCUIT DIAGRAMS





HEX NUT



LOCKWASHER

MOUNTING HARDWARE

FIGURE 1 Case and hardware dimensions and circuit diagrams

(E) denotes changes

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Inches	mm	Inches	mm
.002	0.05	.073	1 85
007	0 18	.216	5 49
010	0.25	.220	5 59
.020	0.38	241	6 12
023	0.58	250	6 35
.031	0.79	.311	7 90
032	0.81	343	8 71
.045	1.14	.372	9 45
.063	1.60	.383	9 73
		921	23.39

NOTES.
1. Dimensions are in inches
2. Metric equivalents are given for general information only
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm).
4 Circuit diagrams are for information only.
5. All filters shall be supplied with mounting hardware (hex nut and lockwasher) Mounting hardware shall be furnished with the same finish as the filter case
6 Imperfect thread or undercut optional, .040 inch (1.02 mm) maximum.
7 One imperfect thread allowed .040 inch (1 02 mm) maximum
8. Recommended mounting torque 64 oz-in ±4 oz-in
9 Turret head terminal optional

FIGURE 1. Case and hardware dimensions and circuit diagrams - Continued.

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REQUIREMENTS.

Design and construction:

Dimensions and configuration: See figure 1.

Weight 2.5 grams maximum

(E) Case finish.

Class B filters, S only (silver plated)

Class S filters, S or G only (silver plated or gold plated) In accordance with MIL-F-28861.

Terminals Solderable

Operating temperature range -55°C to +125°C.

Rated voltage See table I

Rated current 10 amperes maximum

Capacitance to ground See table I for capacitance value

Dissipation factor 3.0 percent maximum

Voltage and temperature limits of capacitance +15 percent, -40 percent

Insulation resistance

At +25°C. 1,000 megohm-microfarads or 100,000 megohms minimum whichever is less

At +125°C. 100 megohm-microfarads or 10,000 megohms minimum, whichever is less

Insertion loss

At +25°C: In accordance with table I

At -55°C and +125°C. A 3 dB degradation from the +25°C value shall be allowed

Voltage drop 0.1 volt maximum

DC resistance. 0.01 ohm maximum

(E) Seal In accordance with MIL-F-28861. Leakage rate for class S filters shall not exceed 1 x 10^{-7} atm cm^3/s.

Temperature rise +25°C maximum

Solderability of terminals. In accordance with MIL-F-28861.

(E) Quality assurance provisions. In accordance with MIL-F-28861

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(E) TABLE I. Electrical characteristics.

Dash number	Circuit	Product assurance level		Rated voltage		Capacitance (µF) -0, +100 percent	Minimum insertion loss (dB) in accordance with MIL-STD-220 1/ 2/					
		Class		DC volts	AC volts 3/		1 MHz	10 MHz	100 MHz	200 MHz	1 GHz	10 GHz
		B	S									
001	P1	X	---	50	---	12	20	38	70	70	70	70
002	C	X	---	50	---	20	26	39	52	60	70	70
003	C	X	X	100	---	05	15	35	42	50	70	70
004	C	X	X	100	---	.10	20	38	48	53	70	70
005	P1	X	---	100	---	075	18	38	70	70	70	70
006	C	X	X	200	115	1,000 pF	---	4	20	25	40	50
007	C	X	X	200	115	5,000 pF	---	15	34	41	50	55
008	C	X	X	200	115	01	4	21	35	40	55	60
009	P1	X	X	200	---	.012	---	12	60	70	70	70
010	C	X	X	200	115	.025	8	28	36	44	64	70

1/ For C circuits, insertion loss measurements shall be made under no load For P1 circuits, insertion loss measurements shall be made under full load over the frequency range of 1 MHz to 10 MHz, above this frequency range, insertion loss measurements shall be made under no load

2/ The insertion loss requirements between any two adjacent specified frequencies shall be that of the lower of the two frequencies in order to accommodate resonant dips

3/ 0 to 400 Hz

Product assurance level: See table I

Part or Identifying Number (PIN). The PIN shall be as follows:

M28861/10 - XXX X X

- XXX: Dash no per figure 1 and table I
- X: Case finish
- X: Product assurance level per MIL-F-28861 and table I

Marking Filters shall be marked, as a minimum, with an abbreviated PIN as shown on figure 2 Full marking, in accordance with MIL-F-28861, shall be marked on the unit package

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J - Abbreviation for JAN
28861 - Abbreviation for M28861
E - Indicates specification sheet 10
C - Code indicating dash no as follows

Code	Dash no	Code	Dash no
A	001	F	006
B	002	G	007
C	003	H	008
D	004	J	009
E	005	K	010

B - Code indicating product assurance level (for example, B indicates class B)
221 - Date code First digit indicates the year (for example, 2 indicates 1992) The last two digits indicate the week

FIGURE 2 Example of marking for the abbreviated PIN on the hex flats - expanded view

Cataloging information Dash numbers 002 through 004, 006 through 008, and 010 shall be cataloged under FSC 5910 as feed-through ceramic capacitors. Dash numbers 001, 005, and 009 shall be cataloged under FSC 5915 as radio frequency interference filters

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities
Army - AR, MI
Air Force - 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - AS, CG, MC, OS, SH
Air Force - 19

Preparing activity:
Navy - EC

Agent
DLA - ES

(Project 59GP-0119-03)

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